

Seongsu Kim

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Ph.D. Student

Graduate School of Artificial Intelligence

 [seongsukim-ml.github.io](https://github.com/seongsukim-ml)

 seongsu.kim@kaist.ac.kr

RESEARCH INTEREST

My research interests include integrating AI into scientific research and using AI to uncover scientific facts. Additionally, I am interested in physical and chemical concepts such as **solid state physics** [C1], **molecular science** [C1,C4,C5], **density functional theory (DFT)** [C1,C4,C5], and **structure prediction** and **material generation** on structured materials like MOF [C2,C3,P1].

EDUCATION

9/2025 - current **Ph.D. Student, Korea Advanced Institute of Science and Technology (KAIST), Korea**
Graduate School of Artificial Intelligence
Advisor: Sungsoo Ahn

2/2023 - 8/2025 **M.S., Pohang University of Science and Technology (POSTECH), Korea**
Graduate School of Artificial Intelligence
Advisor: Sungsoo Ahn, and Dongwoo Kim

3/2016 - 2/2023 **B.S., Gwangju Institute of Science and Technology (GIST), Korea**
Majored in *Physics*
Minored in *Mathematics, Computer Science, Artificial Intelligence*

7/2017 - 8/2017 **University of California, Berkeley**
Summer study abroad program

PUBLICATIONS

[C] Conference
[P] Preprint
† Corresponding

[C5] Machine Learning Hamiltonians are Accurate Energy-Force Predictors

Seongsu Kim, Chanhui Lee, Yoonho Kim, Seongjun Yun, Honghui Kim, Nayoung Kim, Changyong Park, Sehui Han, Sungbin Lim[†], and Sungsoo Ahn[†]
International Conference on Machine Learning (ICML), 2026, [PDF](#)

[P2] MADField: Multi-fidelity Amortized Density Field for Adsorption in Nanoporous Materials

Yoonho Kim, [Seongsu Kim](#), Sungsoo Ahn[†], and Honghui Kim[†]
Preprint 2026, [PDF](#)

[P1] ATOMMOF: All-Atom Flow Matching for MOF-Adsorbate Structure Prediction

Nayoung Kim, Honghui Kim, Sihyun Yu, Minkyu Kim, [Seongsu Kim](#), and Sungsoo Ahn[†]
Preprint 2026, [PDF](#)

[C4] High-order Equivariant Flow Matching for Density Functional Theory Hamiltonian Prediction

[Seongsu Kim](#), Nayoung Kim, Dongwoo Kim[†], and Sungsoo Ahn[†]
Neural Information Processing Systems (NeurIPS), 2025, [PDF](#) [CODE](#) [SLIDE](#)
Spotlight (3.1%~688/21575)

[C3] Flexible MOF Generation with Torsion-Aware Flow Matching

Nayoung Kim, [Seongsu Kim](#), and Sungsoo Ahn[†]
Neural Information Processing Systems (NeurIPS), 2025, [PDF](#)

[C2] MOFFlow: Flow Matching for Structure Prediction of Metal-Organic Frameworks

Nayoung Kim, [Seongsu Kim](#), Minsu Kim, Jinkyu Park[†], and Sungsoo Ahn[†]
International Conference on Learning Representations (ICLR), 2025, [PDF](#)
NeurIPS AIDrugX Workshop, 2024

[C1] Gaussian Plane-wave Neural Operator for Electron Density Estimation

[Seongsu Kim](#), and Sungsoo Ahn[†]
International Conference on Machine Learning (ICML), 2024, [PDF](#) [CODE](#)

EXPERIENCE

2/2025 - current **Structure and Probabilistic Machine Learning (SPML) Lab, Korea**
KAIST, Korea Advanced Institute of Science and Technology (Advisor: Prof. Sungsoo Ahn)
2/2023 - 2/2025 **POSTECH**, Pohang University of Science and Technology
• Machine learning for Scientific Research
• Project 1: Accelerating the Density Functional Theory
• Project 2: Designing Metal-Organic Framework

Student researcher

9/2021 - 2/2023	Computational Many-body Physics (CMBP) Lab, Korea GIST , Gwangju Institute of Science and Technology (Advisor: Prof. Donghee Kim) • Computer-simulated thermodynamics of <i>solid states physics</i> • Investigated the phase transition of physical models using the Monte Carlo method • Investigated the critical phenomena in the 2D long-range antiferromagnetic Ising model with anisotropy • Wrote the simulation code with C++, MPI and CUDA programming	Research Intern
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6/2022 - 7/2022	Statistical Artificial Intelligence (SAIL) Lab, Korea KAIST , Korea Advanced Institute of Science and Technology (Advisor: Prof. Jaesik Choi) • Investigated the various techniques of <i>explainable A.I.</i> including LIMEs, LRP, CRP, and GRAD-CAM.	Research Intern
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12/2019 - 2/2020	Quantum Field & Gravity Theory Group, Korea GIST , Gwangju Institute of Science and Technology (Advisor: Prof. Keunyoung Kim) • Investigated the correspondence of deep learning and the Ads/CFT	Research Intern
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TALKS & PRESENTATION

10/31/2025	QHFlow: Accelerating DFT with Equivariant Flow Matching NVIDIA BioNeMo Team (Hosted by Youhan Lee) SLIDE	Invited Talk
5/6/2025	Accelerating the <i>ab-initio</i> Calculation with the Machine Learning KAIST-MILA Prefrontal AI Research Center	Invited Talk
15/7/2024	Gaussian Plane-wave Neural Operator for Electron Density Estimation KAIST-POSTECH joint AI Workshop	Presentation

HONORS & AWARDS

2016 - 2023	Government-Sponsored Tuition Scholarship Received scholarship 8 times (\$20,000)	Scholarship
2016 - 2017	Government-Sponsored Presidential Science Scholarship Received scholarship 2 times (\$1,400)	Scholarship

REVIEWER

NeurIPS (2025), ICLR (2026, 2025 [Notable reviewer](#)), ICML (2026 [Gold reviewer](#), 2025), AAAI (2023)

WORK EXPERIENCE

1/2020 - 8/2021	Republic of Korea Army, Korea Mandatory military service
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LANGUAGES

English - Professional Working, **Korean** - Native

SKILLS

Backgrounds	Computational physics, Quantum mechanics, Statistical physics
Languages	Python (Proficient) , C++, C, Java
Python Libraries	PyTorch, Lightning, Hydra, PyG, WandB, Numpy, Scikit-learn, Matplotlib
Softwares, etc.	Version control (Git and GitHub), Linux-based environment, Vim, Slurm, Docker
DFT tools	PySCF, ORCA, VASP, Quantum Espresso, Castep
CSP tools	GULP, USPEX, CrySPY